MIL-C-43473F
29 March 1988
SUPERSEDING
MIL-C-43473E
27 December1984

MILITARY SPECIFICATION

Cloth, Coated, Nylon, Polyurethane Coated

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of polyurethane coated rip-stop nylon cloth.

1.2 Classification. The coated cloth shall be of the following types (see 6.2).

Type I - Coated with polyurethane pigmented Olive Green 207
Type II - Deleted
Type III - Woodland camouflage printed and clear polyurethane coated

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

O-I-503 - Insect Repellent, Clothing and Personal Application
PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-C-7020 - Cloth Parachute, Nylon

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1516 - Woodland Pattern - 48 inches
2-1-1516B - Woodland Pattern - 60 inches

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies should be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The coated cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to the characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

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3.3 Materials.

3.3.1 Base cloth.

3.3.1.1 Type I. The base cloth shall be 1.6 ounces per square yard, rip-stop nylon cloth, color Olive Green 106, conforming to type III of MIL-C-7020.

3.3.1.2 Type III. The base cloth shall be 1.6 ounces per square yard, rip-stop nylon cloth conforming to type III of MIL-C-7020. The base cloth shall be dyed to ground shade approximating Light Green 354, and overprinted on one side only by using an acid dyestuff combination. The cloth shall be overprinted using the Woodland camouflage pattern obtained by roller or screen printing using four rollers or screens, as appropriate, for the Light Green 354, Dark Green 355, Brown 356 and Black 357 areas of the pattern (see 6.8).

* 3.3.2 Coating compound. The coating compound for type I coated cloth shall be a composition of polyurethane suitably compounded and pigmented to meet the color requirements in 3.4.1. The coating compound for type III coated cloth shall be a composition of unpigmented polyurethane suitably compounded with a dulling agent to produce a clear, dull coating. For all types of coated cloths, the use of plasticizers is prohibited (see 4.4.1.1 and 6.5). All types of cloth shall be given an approved polyurethane coating (see 6.5).

* 3.3.3 Dusting powder. If dusting powder is used for application on the polyurethane coating, it shall be whiting, talc, or other finely divided mineral material which does not support mildew growth (see 4.4.1.1). Talc shall not contain asbestos and whiting should contain less than 1 percent quartz.

* 3.4 Coated cloth (both types). The cloth shall be coated on one side only with the coating compound specified in 3.3.2 and shall conform to the requirements of table I when tested as specified in 4.4.3. The type III cloth shall be coated on the printed side. For both types, the use of an acrylic anchor coat is permissible. A suitable, approved water repellent finish (see 6.6) shall be applied to the uncoated side of the coated cloth to ensure conformance to the water wicking requirements of the finished coated cloth. At the option of the contractor, the coated side may be dusted with dusting powder specified in 3.3.3.

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TABLE I. Coated cloth requirements

Characteristic	Requirement Minimum	Maximum
Weight, oz/sq yd	2.50	3.25
Breaking strength, pounds:		
Warp	50	--
Filling	50	--
Tearing strength, grams:		
Warp	700	--
Filling	900	--
Hydrostatic resistance, psi:		
Initial	60	--
After accelerated weathering	25 1/	--
After abrasion	35	--
After strength of coating	35	--
After high humidity	40 1/	--
Colorfastness after accelerated weathering	No appreciable change 2/	--
Colorfastness after high humidity	No appreciable change 2/	--
Stiffness, warp only, mm:		
At standard conditions	--	70
At 0°F	--	100
Adhesion of coating, lbs per 2-inch width	7.0	--

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TABLE I. Coated cloth requirements - Continued

Characteristic	Requirement Minimum	Maximum
Blocking, scale rating	--	(2)
Water wicking, inches	--	1/8
Resistance to leakage	No leakage 3/	--
Water absorption, percent	--	7.5
Resistance to insect repellent	4/	--

1/ The coating shall not become stiff and brittle nor soft and tacky and there shall be no evidence of cracking or crazing under visual examination.

2/ "Appreciable change in color" means a change which is immediately noticeable in comparing the test specimen with the original sample for comparision. If closer inspection or a change of angle of light is required to make apparent a slight change of color, the change is not considered appreciable.

3/ Leakage is defined as the appearance of water anywhere within the 4 1/2 inch diameter test area.

4/ The coated cloth shall show no lifting of the coating, no tackiness, no flowing of coating, no pickoff of the coating, no adherence of the coating to itself greater than scale rating No. 3 (slight blocking).

3.4.1 Color of coated side.

3.4.1.1 Type I. The coated side shall be Olive Green 207 and shall match the standard sample (see 6.3).

3.4.1.2 Type III. The finished coated cloth shall match shade Light Green 354, Dark Green 355, Brown 356, and Black 357, each area matching the specific colors of the pattern in accordance with applicable reference standard.

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* 3.4.1.3 Colorfastness to crocking.

a. For type I, applicable to coated side of cloth

b. For type III, applicable to uncoated side of cloth

The coated and printed finished cloth shall show fastness to crocking equal to or better than the standard sample in respect to the solid shades for types I and III cloth. When a standard sample has not been established or designated as applicable to colorfastness, all colors in the print pattern, except Black 357, and the solid shade (type III) shall show an AATCC Chromatic Transference Scale rating for crocking not less than 3.5. Black 357 shall show an AATCC Chromatic Transference Scale rating for crocking not less than 1.5 when tested as specified in 4.4.3.

* 3.4.1.4 Matching. The finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.4.1.5 Pattern execution (type III). The pattern shall reproduce the standard sample in respect to design, colors, and registration of the respective areas. The pattern repeat of the dyed, printed, and finished cloth shall be 27.25 inches + 1.25 inches -2.50 inches, in the warp direction. The various areas of the pattern shall be properly registered in relation to each other and present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided, and the pattern on the finished cloth shall match that of Drawing 2-1-1516 or 2-1-1516B, as applicable (see 6.2).

3.4.1.6 Infrared reflectance, type I. The coated side of the cloth shall have an infrared reflectance of 21 ± 5 percent measured at 1 micron when tested as specified in 4.4.3.

3.4.1.7 Infrared reflectance, type III. The infrared reflectance of the colors in the four-color camouflage printed finished cloth shall conform to the requirements specified in table II when tested as specified in 4.4.3 (see 6.8).

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TABLE II. Infrared requirements, type III

Wavelength nanometers	Reflectance values (percent) Black 357 min	Black 357 max	Light Green 354 min	Light Green 354 max	Dark Green 355 and Brown 356 min	Dark Green 355 and Brown 356 max
600	-	10	8	20	3	13
620	-	10	8	20	3	13
640	-	10	8	20	3	13
660	-	10	8	22	3	13
680	-	10	10	36	3	22
700	-	10	18	60	8	46
720	-	10	26	78	20	68
740	-	10	40	90	30	81
760	-	10	50	92	32	88
780	-	10	55	92	32	90
800	-	10	55	92	32	91
820	-	10	55	92	32	92
840	-	10	55	92	32	92
860	-	10	55	92	32	92

3.5 Length and put-up. The coated cloth shall be put-up on rolls as specified in 5.1. The minimum length of any one roll shall be 50 yards. The maximum number of pieces per roll shall be three, and no single piece shall be less than 25 yards. The ends of the pieces shall be overlapped, not joined by a seam.

3.6 Width. The minimum overall width, after selvage trimming, shall be as specified (see 6.2). Selvages shall be trimmed to give a straight, uniform edge.

3.7 Odor. The coated cloth shall be free from any objectionable odor. Characteristic odor of coating is not considered objectionable.

3.8 Workmanship. The finished coated cloth shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the specified acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and shall be tested for the characteristics specified in 4.4.3 and table IV. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.1.1 Certification. Components and materials listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements:

a. The coating compound is polyurethane, and no plasticizer has been used.

b. The dusting powder used does not support mildew growth, (when coated cloth is dusted).

c. A suitable water repellent finish has been applied to the uncoated side of the coated cloth.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The coated cloth shall be examined for the defects in table III. The same yardage shall be inspected for pinholes and thinly coated areas in accordance with the through-light inspection of MIL-STD-1487. The through-light inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The number of rolls from which the sample yardage is to be selected shall be in accordance with table IV. The sample yardage shall be apportioned equally among the selected rolls.

TABLE III. Yard-by-yard examination defects

Examine	Defect
Coating	Any cut, hole, tear, scratch, or abrasion mark
	Any pinhole
	Any uncoated area
	Any thinly coated area
	Any pit, blister, tunnel, or delamination of coating
	Any lump or heavily coated area

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TABLE III. Yard-by-yard examination defects - Continued

Examine	Defect
Coating - continued	Any crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure
	Any spot, stain, or streak more than 1 inch in combined directions 1/
	Any embedded foreign matter
	Any scorch or burn
	Any piece coated on both sides
Camouflage pattern (type III only)	Any skitteriness of pattern exceeding that shown by the standard sample
	Pattern design not equal to standard sample
	Excessive feathering or spew of pattern
	Pattern repeat not equal to the standard sample
	Warpwise pattern repeat less than 24.75 inches or more than 28.50 inches
Width	Trimmed width less than minimum specified
	Selvages not trimmed
	Edges not straight and uniform
General	Objectionable odor 2/
	Not clean
	Excess of dusting powder, not easily removable by washing
	Curled, folded, scalloped, or rolled edges

1/ Clearly visible at normal inspection distance (approximately 3 feet).

2/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

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TABLE IV. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number 2/
Up to 1,200 inclusive 1/	3	0
1,201 up to and including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination defects only (see 4.4.2.2).

4.4.2.2 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table IV or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

Any roll containing more than three pieces.
Any piece in roll less than 25 yards in length.
Any roll with a total length of less than 50 yards.
Any roll with a total length more than 2 yards less than that marked on ticket.
Any piece joined by a seam.

4.4.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. If any roll in the sample is off shade or shaded side to side, side to center, or end to end, or if any roll does not have the same appearance as the standard sample, the lot represented by the sample shall be rejected.

4.4.3 End item testing. The coated cloth shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed.

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The sample unit shall be 3 continuous yards full width of the coated cloth. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values used in expressing the final results. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

* 4.4.3.1 Standard test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191, ASTM D 618 (Procedure A), or FED-STD-601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method
Weight	3.4	5041
Breaking strength	3.4	5102
Tearing strength	3.4	ASTM D 1424
Hydrostatic resistance:		
Initial	3.4	5512 1/
After accelerated weathering	3.4	5804 2/ and 5512 1/
After abrasion	3.4	5302 3/ and 5512 1/
After strength of coating	3.4	5972 4/ and 5512 1/
After high humidity	3.4	4.5.1 and 5512 1/
Colorfastness after accelerated weathering	3.4	5804 2/

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TABLE V. End item tests - Continued

Characteristic	Requirement paragraph	Test method
Colorfastness after high humidity	3.4	4.5.1
Stiffness, warp only:		
At standard conditions		3.45204
At 0° ± 2°F	3.4	5204 5/
Adhesion of coating	3.4	5970
Blocking	3.4	5872 6/
Water wicking	3.4	4.5.2
Resistance to leakage	3.4	4.5.3
Water absorption	3.4	5502 7/
Resistance to insect repellent	3.4	4.5.4
Colorfastness to crocking:		
Type I (coated side)	3.4.1.3	5651
Type III (uncoated side)	3.4.1.3	5651
Infrared reflectance:		
Type I	3.4.1.6	4.5.6.1
Type III	3.4.1.7	4.5.6.1

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1/ If leakage occurs under 10 psi on any specimen, that reading shall be discarded and another specimen shall be tested. If leakage occurs under 10 psi on more than one of the five specimens, all readings shall be valid. The water pressure shall be applied only to the uncoated side of the coated cloth.

2/ The coated side of the cloth shall be exposed to the light source. The exposure period in the weatherometer shall be 100 hours. At the end of the exposure period, the specimens shall be visually examined to determine if the coating has become stiff and brittle or soft and tacky, or if there is any evidence of cracking or crazing, and then tested for hydrostatic resistance.

3/ A solid rubber diaphragm 0.030 inch ± 0.010 inch (no metallic contact) shall be used. The abradant shall be the coated side of the coated cloth. The coated side of the specimen shall be abraded 1000 double strokes and then tested for hydrostatic resistance with the abraded portion of the test specimen centered in the hydrostatic test area.

4/ Except that the specimens shall be stretched at 20 pounds.

5/ The test specimens shall be subjected to a temperature of 0° ± 2°F for a minimum of 4 hours, and the test shall be performed in a still atmosphere at that temperature.

6/ Except that only one specimen shall be tested.

7/ Except that all weighings shall be made to the nearest 0.01 gram.

4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Resistance to high humidity test. Three 4-inch by 4-inch specimens shall be laid flat, coated side up on supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1 inch below the specimens. The lid of the desiccator shall be put in place and the dessicator placed in a circulating air oven having a temperature of 180° ± 2°F for a period of 14 days. At the end of the aging period, each specimen shall be removed from the desiccator, visually examined for colorfastness and then tested immediately in accordance with Method 5512 of FED-STD-191 with the water pressure being applied to the uncoated side.

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4.5.2 Water wicking test. One 2 inch by 7 inch sample of coated cloth with the long dimension in the filling direction shall be the test specimen. One end of the specimen shall be weighted (10 to 20 grams) across the full width of the specimen and immersed to a depth of 2 inches in distilled water at room temperature for 5 minutes. When withdrawn from the water, the height of the wetted area above the water line (on the uncoated side of the specimen) shall be immediately measured to the nearest 1/32 inch. Three determinations shall be made per sample unit.

4.5.3 Resistance to leakage test. The test for resistance to leakage shall be conducted as specified in Method 5516 of FED-STD-191 except that the uncoated side of the cloth shall contact the water. The hydrostatic head shall be 50 centimeters and shall be held for 10 minutes. The report shall only include "measurement of the appearance of water drops."

4.5.4 Resistance to insect repellent test. The insect repellent shall conform to type II, concentration A, of O-I-503. Three drops of the repellent shall be placed in the center of a 4 inch by 8 inch sample of coated cloth with the repellent contacting the coated side. The sample shall be folded to form a 4 inch by 4 inch square with the coated sides contacting each other. The folded sample shall then be placed between two 6 inch by 6 inch glass plates and a 4 pound weight placed on the assembly and left at standard conditions for 16 hours. The sample shall then be removed from between the glass plates, rated for blocking as required in Method 5872 of FED-STD-191, and immediately examined for conformance to the requirements in table I.

4.5.5 Colorfastness to crocking (dry only) test. Colorfastness to crocking shall be tested as specified in Method 5651 of FED-STD-191. When testing for colorfastness to crocking, the individual shades of the pattern shall be so selected that only the specific shade being evaluated is included in the test area of the submitted specimen as well as of the standard. Individual shades in the pattern shall be evaluated and reported separately.

4.5.6 Infrared reflectance tests.

4.5.6.1 Type I. The infrared reflectance shall be determined on the coated side of the cloth by recording the spectral reflectance relative to MgO or $BaSO_4$ at 1 micron using a spectrophotometer sensitive in this range (see 6.7). Readings shall be made on two layers of the specified color while backed with cloth, cotton, sateen, OG-107, and values shall be reported to the nearest 1 percent. The infrared reflectance value shall be the average of three individual readings.

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4.5.6.2 Type III. Infrared reflectance data shall be determined on the coated side of the cloth and shall be obtained from 600 to 860 nanometers (nm) relative to a barium sulfate standard white reference (see 6.9). The spectral band width at 860 nm shall be less than 25 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer (see 6.7) shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE source A or CIE source D65. Specimens shall be measured as a single layer backed with six layers of the same shade cut from the standard. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent; and the wavelength accuracy shall be within 2 nm. Readings shall be taken on a minimum of two different areas and the data averaged.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The coated cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth is intended for use in the fabrication of ponchos, ground cloths, and components of jungle hammocks.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.2, 4.3 and 6.4).
d. Woodland camouflage pattern drawing if required (see 3.4.1.5).
e. Width of cloth required (see 3.6).
f. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

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6.3 Standard sample and pattern drawing. For access to the standard shade samples (see 3.1) and the Woodland camouflage pattern drawing, if applicable (see 3.4.1.5), address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.5 Coating. The incorporation of an isocyanate (PAPI or MDI) in the top coating using a thermo-plastic polyurethane, or the use of a thermo-setting or cross linked polyurethane has been found necessary to meet the requirements of this document. The coating limitations are intended to guarantee a waterproof cloth with adequate resistance to mildew and bacterial degradation (see 3.3.2). Approval of such components and combinations are the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, and is based on more extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement (see 3.3.2).

* 6.6 Water repellents. Water emulsion or solvent soluble silicone water repellents have been found satisfactory for this purpose (see 3.4). Approval of such components and combinations are the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, and is based on more extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement (see 3.4).

* 6.7 Spectrophotometer. Suitable spectrophotometers for measuring spectral reflectance in the visible, near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR, Hunter VIS/NIR Spectrocolorimeter and Macbeth 1500 with IR option (see 4.5.6.1 and 4.5.6.2).

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6.8 Pattern areas. The areas of the pattern for type III have been found to be satisfactory when dyed or printed with combinations of the following colorants:

The following dyes have been found acceptable for dyeing the ground shade approximating Light Green 354:

Acid Blue 258

Acid Orange 4R

Acid Yellow 219

Printing of the Light Green 354, Dark Green 355, and Brown 356 shades can be accomplished by varying the amount of the following dyestuffs:

Acid Orange 156

Acid Red 266

Acid Blue 258

Shade Black 357 may also be satisfactorily printed with the use of carbon black alone or in combination with the above dyes.

* 6.9 Source of material. Barium sulfate of suitable quality for use as a white standard is available from Eastman Kodak Co., Rochester, NY 14600.

* 6.10 Subject term (key word) listing.

Cloth, nylon, polyurethane coated
Ground cloths
Ponchos

6.11 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicated where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Navy - NU
Air Force - 99

Review activities:

Army - MD, ME
Navy - MC
Air Force - 82
DLA - CT
DNA - DS

Preparing activity:

Army - GL

Project No. 8305-0181

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-43473F	Cloth, Coated, Nylon, Polyurethane Coated

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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